Database Part – 1

Folder Structure:

```
L9
  .venv
  a1
  lab9
    __pycache__
    __init__.py
    asgi.py
    settings.py
    urls.py
    wsgi.py
  q1
    __pycache__
    migrations
    templates / q1
      index.html
    __init__.py
    admin.py
    apps.py
    forms.py
    models.py
    tests.py
    urls.py
    views.py
  q2
  db.sqlite3
  manage.py
```

Settings.py:

```
INSTALLED_APPS = [
'django.contrib.admin',
'django.contrib.auth',
'django.contrib.contenttypes',
'django.contrib.sessions',
'django.contrib.messages',
'django.contrib.staticfiles',
'q1',
'q2',
'a1',
```

```
]

DATABASES = {
'default': {
'ENGINE': 'django.db.backends.sqlite3',
'NAME': BASE_DIR / 'db.sqlite3',
}
}
```

Urls.py:

```
from django.contrib import admin
from django.urls import include, path

urlpatterns = [
path('admin/', admin.site.urls),
path('q1/', include('q1.urls')),
path('q2/', include('q2.urls')),
path('a1/', include('a1.urls')),
]
```

A1:

Admin.py:

```
from django.contrib import admin

from .models import Institute

admin.site.register(Institute)
```

Models.py:

```
from django.db import models

class Institute(models.Model):
institute_id = models.AutoField(primary_key=True)
name = models.CharField(max_length=150)
no_of_courses = models.PositiveIntegerField()

class Meta:
db_table = 'Institutes'

def __str__(self):
return self.name
```

Urls.py:

```
from django.urls import path
```

```
from . import views

app_name = 'a1'

urlpatterns = [
path('', views.index, name='index'),
]
```

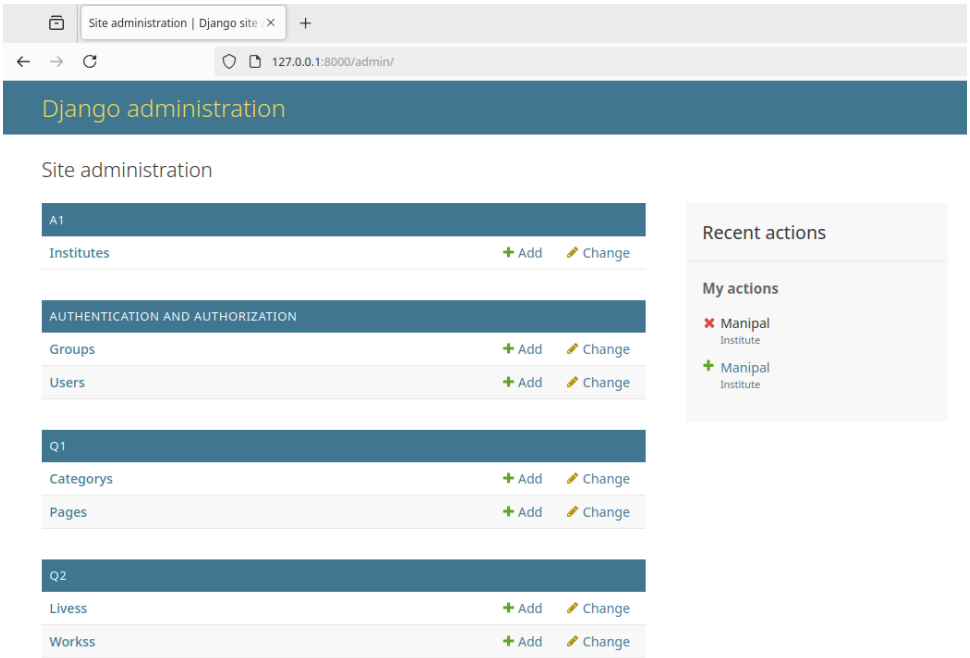
Views.py:

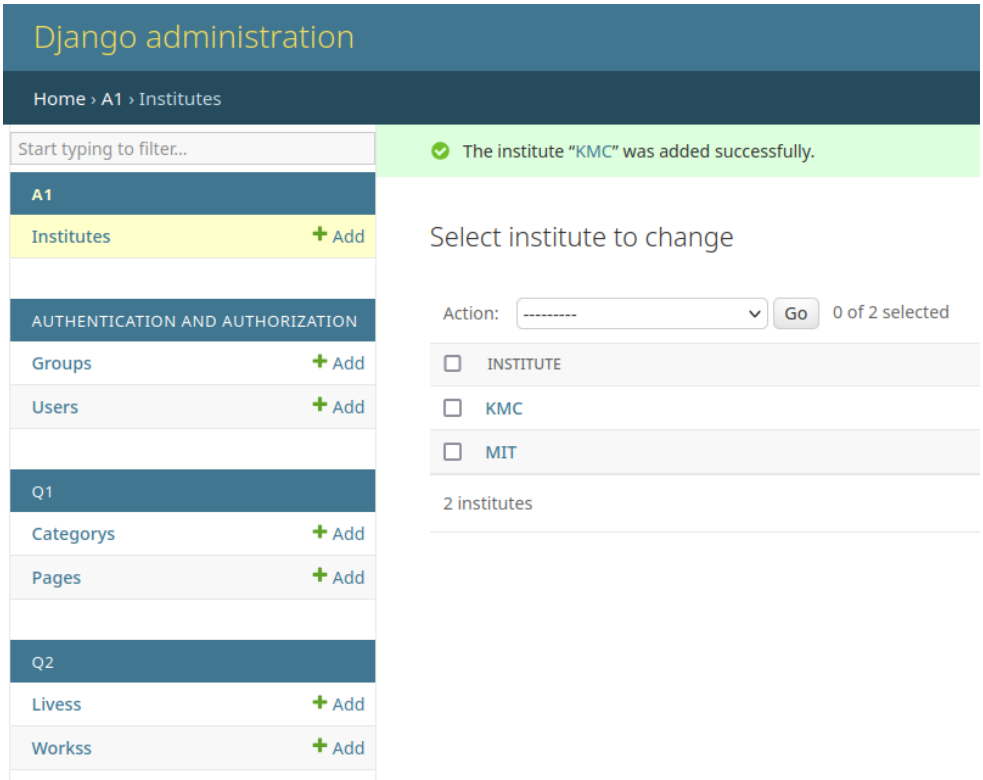
```
from django.shortcuts import render

from .models import Institute



def index(request):
institutes = Institute.objects.all().order_by('name')
return render(request, 'a1/index.html', {'institutes': institutes})
```

Index.html:

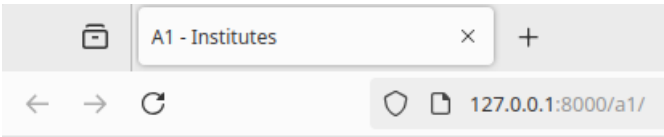
```
<!DOCTYPE html>
<html>
<head>
<title>A1 - Institutes</title>
</head>
<body>
<h1>A1: Institutes</h1>

<label for="institute_list">Institute Names</label><br>
<select id="institute_list" size="8">
{% for institute in institutes %}
<option>{{ institute.name }}</option>
{% empty %}
<option>No institutes found.</option>
{% endfor %}
</select>
</body>
</html>
```

Screenshots:


Site administration | Django site
127.0.0.1:8000/admin/
Django administration
Site administration
A1
Institutes
Add
Change
AUTHENTICATION AND AUTHORIZATION
Groups
Users
Q1
Categorys
Pages
Q2
Livess
Workss
Recent actions
My actions
Manipal
Institute
Manipal
Institute


Django administration
Home › A1 › Institutes
Start typing to filter...
A1
Institutes
Add
AUTHENTICATION AND AUTHORIZATION
Groups
Users
Q1
Categorys
Pages
Q2
Livess
Workss
The institute "KMC" was added successfully.
Select institute to change
Action:

Go
0 of 2 selected
INSTITUTE
KMC
MIT
2 institutes



A1: Institutes

Institute Names

KMC
MIT